## FinBERT Quantum Optimization — Deliverables

### Reproducible Code

All scripts are located in the project's src/ directory and can rebuild the full pipeline end to end.

The execution order is:

prepare_data.py

> Loads raw CSV and AllAgree files, cleans data, maps labels (negative=0, neutral=1, positive=2), performs stratified 70/15/15 split with seed=42, saves train/val/test CSVs for both tracks.

build_qubo.py

> Trains FinBERT with all 32 hyperparameter combinations, records validation loss for each, constructs the 5×5 QUBO matrix via least-squares regression with baseline subtraction.

run_sa.py

> Runs Simulated Annealing (10 runs, 500 iterations, geometric cooling) on the QUBO matrix.

run_qaoa.py

> Runs QAOA at depths p=1,2,3 using Qiskit StatevectorSampler and COBYLA optimizer.

run_vqe.py

> Runs SamplingVQE with RealAmplitudes and EfficientSU2 ansatze, COBYLA and Nelder-Mead optimizers.

train_baseline.py

> Trains FinBERT with a given config, evaluates on test set, saves metrics JSON, loss plot, and best model checkpoint.

evaluate_optimized.py

> Calls train_baseline with baseline, QUBO-optimized, and sweep-best configs for both tracks.

Environment: Python 3.13, Qiskit 2.3.1, qiskit-ibm-runtime 0.45.1, qiskit-algorithms, qiskit-optimization, PyTorch, Hugging Face transformers, scikit-learn, pandas, matplotlib.

Base checkpoint: ahmedrachid/FinancialBERT (pre-trained on financial corpora, no sentiment fine-tuning)

Tokenizer: ahmedrachid/FinancialBERT, max_length=64

Random seed: 42 (applied to data split, torch.manual_seed, numpy.random.seed)

Label mapping: negative=0, neutral=1, positive=2

Quantum execution: All runs on local noiseless simulation via Qiskit Aer. No IBM Quantum hardware was used.

Note on versioning: A progress update sent July 1 reported baseline accuracy of 84.3% (primary) and 97.1% (AllAgree). These were generated before a dropout implementation bug was corrected. The final results in this report reflect a complete re-run of all experiments after the fix, ensuring all configurations were trained under identical, correct conditions.

**Metrics Tables**

| **Primary CSV Track (4,846 sentences)**<br>Train: 3,392 \| Validation: 727 \| Test: 727 \| Split: 70/15/15 stratified \| Seed: 42 | | | | | | |
|---|---|---|---|---|---|---|
| **Configuration** | **Hyperparameters** | **Accuracy** | **Precision** | **Recall** | **Macro F1** | **Runtime** |
| Baseline FinBERT | LR=2e-5, DO=0.1, BS=32, EP=3 | 82.4% | 0.79 | 0.84 | 0.81 | 30.3 min |
| FinBERT + SA/QAOA/VQE | LR=1e-5, DO=0.3, BS=16, EP=3 | 80.9% | 0.77 | 0.84 | 0.80 | 35.9 min |
| FinBERT + Sweep Best | LR=5e-5, DO=0.3, BS=32, EP=3 | 82.9% | 0.80 | 0.83 | 0.82 | 31.6 min |

SA/QAOA/VQE all converged to the same hyperparameter configuration. The sweep-best row shows the actual best configuration by validation loss from the 32-combination search, included to separate the QUBO approximation quality from the optimizer performance.

| **Repeated Trials – Primary CSV Track**<br>(5 seeds: 42, 123, 456, 789, 1024, 95% CI) | | |
|---|---|---|
| **Configuration** | **Accuracy (mean ± CI)** | **Macro F1 (mean ± CI)** |
| Baseline FinBERT | 83.2% ± 0.9% | 0.818 ± 0.009 |
| FinBERT + SA/QAOA/VQE | 81.9% ± 0.7% | 0.802 ± 0.005 |
| FinBERT + Sweep Best | 83.6% ± 0.4% | 0.826 ± 0.005 |

Baseline and sweep-best confidence intervals overlap, confirming performance differences fall within run-to-run variance. The QUBO-optimized configuration consistently underperforms the baseline.

| **AllAgree Track — Financial PhraseBank 100% Agreement (2,264 sentences)**<br>Train: 1,584 \| Validation: 340 \| Test: 340 \| Split: 70/15/15 stratified \| Seed: 42 | | | | | | |
|---|---|---|---|---|---|---|
| **Configuration** | **Hyperparameters** | **Accuracy** | **Precision** | **Recall** | **Macro F1** | **Runtime** |
| Baseline FinBERT | LR=2e-5, DO=0.1, BS=32, EP=3 | 97.6% | 0.97 | 0.97 | 0.97 | 15.5 min |
| FinBERT + SA/QAOA/VQE | LR=2e-5, DO=0.1, BS=16, EP=4 | 97.1% | 0.95 | 0.97 | 0.96 | 22.2 min |
| FinBERT + Sweep Best | LR=5e-5, DO=0.1, BS=16, EP=4 | 96.2% | 0.95 | 0.94 | 0.95 | 22.9 min |

All configurations perform within 1.5 percentage points of each other, reflecting a performance ceiling on the unambiguous AllAgree subset.

| **Repeated Trials – AllAgree Track**<br>(5 seeds: 42, 123, 456, 789, 1024, 95% CI) | | |
|---|---|---|
| **Configuration** | **Accuracy (mean ± CI)** | **Macro F1 (mean ± CI)** |
| Baseline FinBERT | 97.4% ± 0.5% | 0.964 ± 0.009 |
| FinBERT + SA/QAOA/VQE | 96.8% ± 0.3% | 0.955 ± 0.006 |
| FinBERT + Sweep Best | 96.1% ± 0.5% | 0.945 ± 0.010 |

All three configurations produce overlapping confidence intervals, consistent with a performance ceiling on the AllAgree subset.

| Reference Paper Benchmark (AllAgree Track) | | | | |
|---|---|---|---|---|
| **Configuration** | **Paper Acc.** | **Paper F1** | **This Study Acc.** | **This Study F1** |
| Baseline FinBERT | 85.0% | 0.81 | 97.6% | 0.97 |
| FinBERT + SA | 87.1% | 0.84 | 80.9%* | 0.80* |
| FinBERT + QAOA | 87.9% | 0.86 | —† | —† |
| FinBERT + VQE | 88.0% | 0.87 | —† | —† |

*SA/QAOA/VQE all found the same config, so a single result is reported. †QAOA and VQE selected the identical configuration as SA; separate test results are not applicable.

| Optimizer Convergence Summary | | | | | |
|---|---|---|---|---|---|
| **Track** | **Method** | **Bitstring** | **QUBO Energy** | **Config Found** | **Consistent** |
| Primary | SA (10/10 runs) | (0,0,1,0,0) | -0.0169 | LR=1e-5, DO=0.3, BS=16, EP=3 | Yes |
| | QAOA (p=1,2,3) | (0,0,1,0,0) | -0.0169 | LR=1e-5, DO=0.3, BS=16, EP=3 | Yes |
| | VQE (4 variants) | (0,0,1,0,0) | -0.0169 | LR=1e-5, DO=0.3, BS=16, EP=3 | Yes |
| AllAgree | SA (10/10 runs) | (0,1,0,0,1) | -0.0334 | LR=2e-5, DO=0.1, BS=16, EP=4 | Yes |

| | | | | | |
|---|---|---|---|---|---|
| | QAOA (p=1,2,3) | (0,1,0,0,1) | -0.0334 | LR=2e-5, DO=0.1, BS=16, EP=4 | Yes |
| | VQE (4 variants) | (0,1,0,0,1) | -0.0334 | LR=2e-5, DO=0.1, BS=16, EP=4 | Yes |

**Diagnostics**

| **Primary Track — Confusion Matrices**<br>Baseline (LR=2e-5, DO=0.1, BS=32, EP=3) — Accuracy: 82.4% | | | |
|---|---|---|---|
| | **Pred: Negative** | **Pred: Neutral** | **Pred: Positive** |
| True: Negative | 82 | 6 | 3 |
| True: Neutral | 24 | 350 | 58 |
| True: Positive | 5 | 32 | 167 |

| **SA/QAOA/VQE Optimized**<br>(LR=1e-5, DO=0.3, BS=16, EP=3) — Accuracy: 80.9% | | | |
|---|---|---|---|
| | **Pred: Negative** | **Pred: Neutral** | **Pred: Positive** |
| True: Negative | 85 | 5 | 1 |
| True: Neutral | 29 | 339 | 64 |
| True: Positive | 11 | 29 | 164 |

| **Sweep Best**<br>(LR=5e-5, DO=0.3, BS=32, EP=3) — Accuracy: 82.9% | | | |
|---|---|---|---|
| | **Pred: Negative** | **Pred: Neutral** | **Pred: Positive** |
| True: Negative | 80 | 6 | 5 |
| True: Neutral | 20 | 364 | 48 |
| True: Positive | 5 | 40 | 159 |

| Primary Track — Training and Validation Loss | | | |
|---|---|---|---|
| **Configuration** | **Train Loss (by epoch)** | **Val Loss (by epoch)** | **Overfitting Pattern** |
| Baseline | 0.554 → 0.271 → 0.144 | 0.422 → 0.434 → 0.499 | Val loss increases from epoch 1 — overfitting |
| SA/QAOA/VQE | 0.690 → 0.408 → 0.329 | 0.546 → 0.483 → 0.536 | High initial val loss, partial recovery then rise — underfitting then overfitting |
| Sweep Best | 0.576 → 0.342 → 0.268 | 0.450 → 0.445 → 0.575 | Stable for 2 epochs then sharp rise — overfitting at epoch 3 |

Failure case notes (Primary): The QUBO-optimized config (LR=1e-5, DO=0.3) regularizes too aggressively. The low learning rate combined with high dropout prevents the model from learning complex patterns in the noisier primary dataset. Neutral-to-Positive confusion increases from 58 (baseline) to 64 (optimized), indicating the model defaults to neutral predictions more often under heavy regularization. The sweep-best config shows the highest Neutral correct predictions (364 vs 350) but increases Neutral-to-Positive confusion (48), suggesting the high learning rate learns faster but less stably.

| AllAgree Track — Confusion Matrices<br>Baseline (LR=2e-5, DO=0.1, BS=32, EP=3) — Accuracy: 97.6% | | | |
|---|---|---|---|
| | **Pred: Negative** | **Pred: Neutral** | **Pred: Positive** |
| True: Negative | 43 | 0 | 3 |
| True: Neutral | 1 | 206 | 2 |
| True: Positive | 0 | 2 | 83 |

| SA/QAOA/VQE Optimized (LR=2e-5, DO=0.1, BS=16, EP=4) — Accuracy: 97.1% | | | |
|---|---|---|---|
| | **Pred: Negative** | **Pred: Neutral** | **Pred: Positive** |
| True: Negative | 46 | 0 | 0 |
| True: Neutral | 4 | 204 | 1 |
| True: Positive | 3 | 2 | 80 |

| Sweep Best (LR=5e-5, DO=0.1, BS=16, EP=4) — Accuracy: 96.2% | | | |
|---|---|---|---|
| | **Pred: Negative** | **Pred: Neutral** | **Pred: Positive** |
| True: Negative | 42 | 0 | 4 |
| True: Neutral | 2 | 206 | 1 |
| True: Positive | 1 | 5 | 79 |

| AllAgree Track — Training and Validation Loss | | | |
|---|---|---|---|
| **Configuration** | **Train Loss (by epoch)** | **Val Loss (by epoch)** | **Overfitting Pattern** |
| Baseline | 0.490 → 0.100 → 0.040 | 0.153 → 0.090 → 0.095 | Minimal — val loss stabilizes low |
| SA/QAOA/VQE | 0.385 → 0.071 → 0.036 → 0.015 | 0.119 → 0.111 → 0.114 → 0.133 | Slight overfitting at epoch 4 |

| | | | |
|---|---|---|---|
| Sweep Best | 0.308 → 0.076 → 0.020 → 0.050 | 0.114 → 0.106 → 0.130 → 0.240 | Clear overfitting at epochs 3-4, train loss spike at epoch 4 |

Failure case notes (AllAgree): All configurations perform at 96-98% accuracy. The few errors are concentrated in Negative↔Positive confusion (misclassifying positive sentences as negative or vice versa), not Neutral confusion. The optimized config achieves perfect Negative recall (46/46) but at the cost of 4 false negatives on Neutral. The sweep-best config overfits significantly by epoch 4, as shown by the validation loss spike to 0.240, yet its test accuracy (96.2%) remains high because the AllAgree dataset contains so few ambiguous sentences.

| **QUBO Surrogate Fit Quality** | | | | | |
|---|---|---|---|---|---|
| **Track** | **$R^2$** | **Spearman ρ** | **Spearman p-value** | **Best Config Match** | **Top-5 Overlap** |
| Primary | 0.67 | 0.85 | <0.001 | No | 2/5 |
| AllAgree | 0.69 | 0.88 | <0.001 | No | 3/5 |

The quadratic surrogate explains approximately two-thirds of the variance in validation loss on both tracks. Spearman rank correlation indicates the QUBO correctly captures the general ordering of configurations (good vs bad) but misranks the top candidates. The predicted best configuration did not match the actual best on either track. This confirms that the QUBO approximation quality — not the performance of SA, QAOA, or VQE — is the primary bottleneck. Potential improvements include averaging validation losses over multiple seeds per configuration to reduce noise in the regression targets, or using a higher-order surrogate model (though this would no longer be a QUBO and could not be solved by standard quantum optimization algorithms).

**Comparison Notes**

Primary CSV Track

The baseline FinBERT achieved 82.4% accuracy and 0.81 macro F1 on the primary CSV test set. Neither the QUBO-optimized configuration (80.9% accuracy, 0.80 F1) nor the sweep-best configuration (82.9% accuracy, 0.82 F1) outperformed the baseline in the reported runs. Repeated trials with 5 random seeds confirmed this: baseline accuracy was 83.2% ± 0.9% and sweep-best was 83.6% ± 0.4% (95% CI), with overlapping confidence intervals.

The QUBO-optimized result underperformed the baseline because the quadratic surrogate favored a high-regularization configuration (LR=1e-5, dropout=0.3) that underfit the data. The QUBO matrix captured the individual benefit of higher dropout and lower learning rate but could not represent the three-way interaction where these settings only work well at certain batch sizes. The sweep-best configuration (LR=5e-5, DO=0.3, BS=32, EP=3) had the lowest validation loss among all 32 combinations but did not reliably outperform the baseline on the test set, likely due to run-to-run variance on a dataset of only 3,392 training samples.

The improvement did not come from quantum optimization, classical tuning, or cleaner data handling. The baseline defaults were already near-optimal for this checkpoint and dataset combination. All three optimizers (SA, QAOA, VQE) correctly solved the QUBO — the limitation was in the QUBO's fidelity as a surrogate for the true validation loss landscape.

Examination of the reference model files revealed the original study used yiyanghkust/finbert-tone, a BERT model already fine-tuned for financial sentiment classification. This study used ahmedrachid/FinancialBERT, a BERT model pre-trained on financial corpora but with no sentiment fine-tuning, as required by the instruction document's directive to build the pipeline independently from raw data. A pre-fine-tuned checkpoint may exhibit greater sensitivity to hyperparameter variation because its classification performance starts lower, creating more room for optimization to demonstrate improvement.

AllAgree Track

The baseline FinBERT achieved 97.6% accuracy and 0.97 macro F1 on the AllAgree test set. The QUBO-optimized configuration achieved 97.1% accuracy (0.96 F1) and the sweep-best achieved 96.2% accuracy (0.95 F1). Neither improved upon the baseline. Repeated trials confirmed all three configurations produce overlapping confidence intervals (baseline 97.4% ± 0.5%, QUBO-optimized 96.8% ± 0.3%, sweep-best 96.1% ± 0.5%).

The AllAgree subset contains only sentences where all 16 human annotators agreed on the sentiment label, making it an inherently unambiguous classification task. Any BERT-class model achieves 95%+ accuracy on this subset, leaving effectively no room for hyperparameter optimization to demonstrate improvement. This ceiling effect was observed across all tested checkpoints (ProsusAI/finbert, ahmedrachid/FinancialBERT with reset head, ahmedrachid/FinancialBERT).

The reference paper reports a baseline of 85.0% on this dataset, which is anomalously low. Their confusion matrix implies a test set of approximately 968 samples rather than the 340 produced by a 15% split of 2,264, suggesting a different split strategy or dataset construction. The paper's incremental staircase of improvement (85% → 87.1% → 87.9% → 88.0%) may reflect training variance rather than genuine differences between optimization methods, given that all three methods found the same hyperparameter configuration.

The improvement did not come from any method. The result is consistent with the expectation that near-ceiling performance cannot be further improved through hyperparameter tuning alone.

**Appendix: Reproducibility Details**

| Parameter | Value |
|---|---|
| Base checkpoint | ahmedrachid/FinancialBERT |
| Tokenizer | ahmedrachid/FinancialBERT, max_length=64 |
| Random seed | 42 (data split, torch, numpy) |
| Data split ratio | 70% train / 15% val / 15% test, stratified |
| Label mapping | negative=0, neutral=1, positive=2 |
| Optimizer | AdamW |
| Primary CSV source | all-data.csv (4,846 sentences) |
| AllAgree source | Financial PhraseBank Sentences_AllAgree.txt (2,264 sentences) |
| Primary CSV preprocessing | CR→LF line ending fix; parsed as label,"text" |
| AllAgree preprocessing | Parsed as text@label (split on last @) |
| QUBO construction | Least-squares regression, 15 params fit to 32 val losses, baseline subtracted |
| SA parameters | 10 runs × 500 iterations, T_init=1.0, cooling=0.995 |
| QAOA | StatevectorSampler, COBYLA (200 iter), depths p=1,2,3 |
| VQE | StatevectorSampler, SamplingVQE, 2 ansatze × 2 optimizers (200 iter each) |
| Quantum hardware | Not used (local simulation only) |

| | |
|---|---|
| Reference paper checkpoint | yiyanghkust/finbert-tone (identified from provided model files) |
| Reference paper batch size | 16 (per trainer_state.json; paper states 32) |
| Repository | https://github.com/fzara007/finbert-quantum-optimization |

File Structure

```
project/
  data/raw/all-data.csv
  data/raw/FinancialPhraseBank-v1.0/Sentences_AllAgree.txt
  data/processed/primary/{train,val,test}.csv
  data/processed/allagree/{train,val,test}.csv
  src/prepare_data.py
  src/train_baseline.py
  src/build_qubo.py
  src/run_sa.py
  src/run_qaoa.py
  src/run_vqe.py
  src/evaluate_optimized.py
  src/repeated_trials.py
  src/qubo_diagnostics.py
  models/{baseline,optimized,sweep_best}_{primary,allagree}/
  outputs/metrics/*.json
  outputs/plots/baseline_loss_*.png
  outputs/metrics/repeated_trials_*.json
  outputs/metrics/qubo_diagnostics_*.json
  docs/FinBert_Model_Deliverables_-_Fiona_Zara.pdf
  docs/FinBert_Model_Summary_-_Fiona_Zara.pdf
```